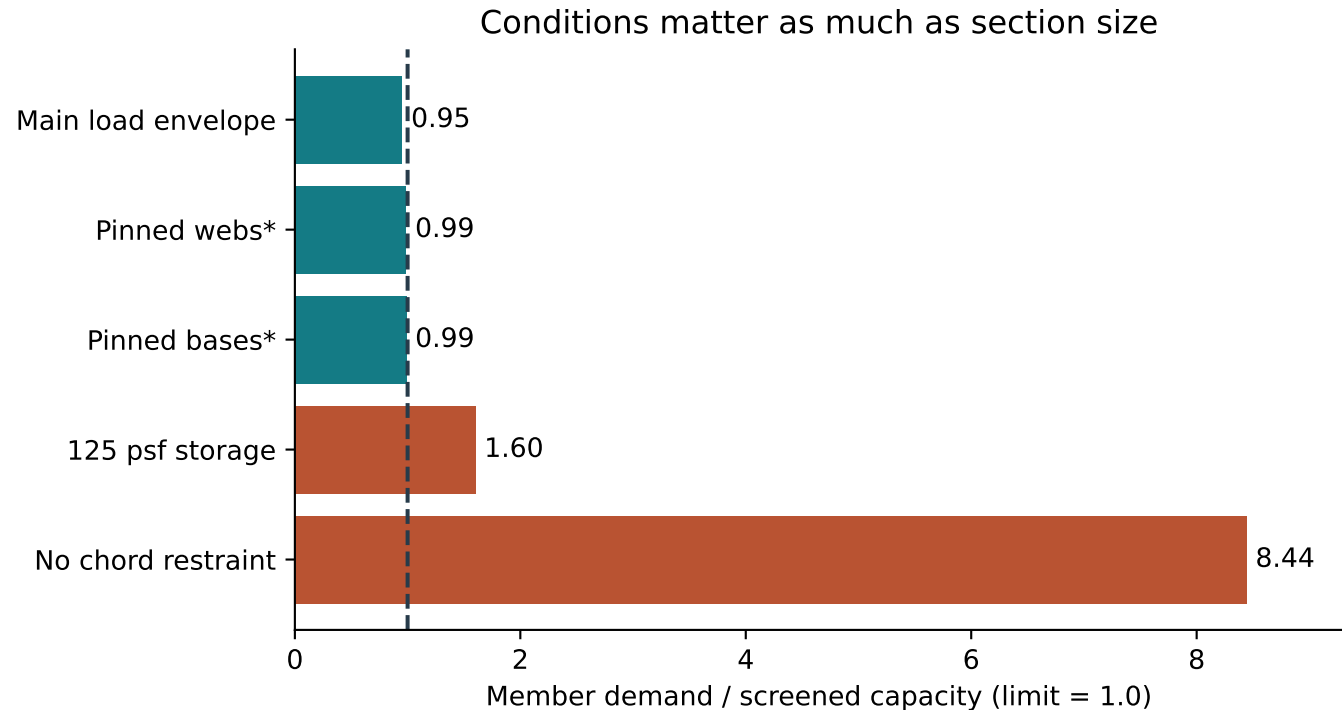
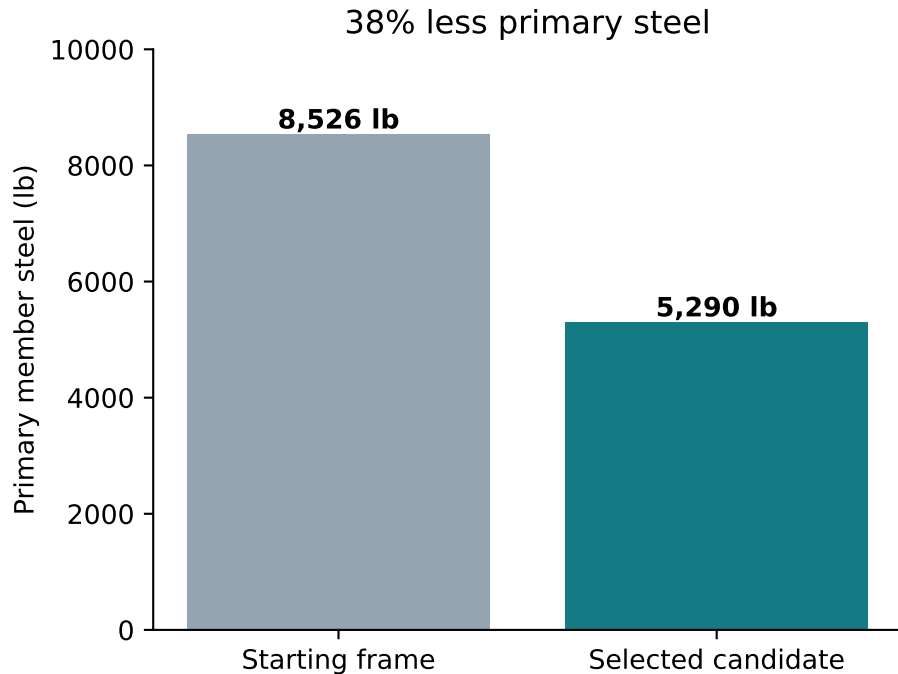


Garage — fixed-geometry member sizing



*Separate sensitivity runs. Requires chord restraint at ≤ 6 ft. Preliminary sizing, not a fabrication design.

Result and scope

The lightest candidate found in this discrete stock search weighs 5,290 lb, compared with 8,526 lb of starting primary steel: a reduction of 3,236 lb (38%). This is a local search result, not proof of the absolute minimum. Geometry, supports and truss arrangement were held fixed.

The audited starting frame exceeds the member screen at W3 (1.065 utilization). Downsizing therefore was not applied uniformly: W3 needs a thicker wall while other members can become lighter. The nine-column layout retains the previously accepted removal of W2; no column was removed in this sizing round.

Main completed and roof-first load envelopes reach 0.951 utilization. Separate pinned-web and pinned-base completed-stage runs reach 0.986 and 0.987. These checks are separate, not a combined pinned-web/pinned-base model. A utilization below 1 means only that the implemented member screen passes.

The result requires positive lateral restraint of chords/rails at no more than 6 ft. Without that restraint the capacity screen fails (8.44). The 125 psf storage sensitivity also fails (1.60). Occupancy classification and the applicable code live load remain unresolved.

Weight excludes secondary roof/floor framing, cladding, PV, actual connection steel and foundations. Their dead loads are budgeted separately. Connections, chord restraint, trolley flange loading, foundations and site-specific wind/seismic design are not complete; these sizes are not a purchase or fabrication schedule.

Audited gravity-load basis

Main loft: 18 ft 8 1/4 in east-west by 14 ft 9 in north-south = 275.64 sq ft. No support capacity is credited to the old walls or slab. Existing garage weight is not assigned to the new independent frame.

Assembly weights below are provisional budgets informed by manufacturer weights, not a completed material takeoff. A product or load-path change requires another solve.

Item	Basis	Applied load / qualification
Main solar roof	10 psf, actual slope area	3 PV/glass + 3 skin + 3 secondary steel + 1 reserve
Rear hip cap	7 psf, actual surface area	3 insulated skin + 3 secondary frame + 1 trim
All roof dead load	Combined surfaces	6,393 lb
Roof live load	20 psf, horizontal projection	13,388 lb; final site requirements pending
Upper metal walls	4 psf, gross surface	2,331 lb; includes panel/girt allowance
Loft floor dead	12 psf, excluding primary steel	3,308 lb; plywood/joists/hardware/reserve
Loft live	40 psf general	11,026 lb
Heavier storage bands	75 psf in approximately 3 ft N/E bands	14,141 lb total floor live in this case
West balcony	13.33 sq ft; 12 dead + 100 live psf	160 lb dead + 1,333 lb live; eccentric onto TW
Machine case	Additional 200 lb	Strength case with the storage-band loading
Moving hoist	1,000 lb rated, one hoist at a time	1.25 impact + 100 lb equipment = 1,350 lb
Primary steel + joints	Catalog weight + 10% allowance	5,819 lb; recomputed at every sizing iteration
Total new dead load	All above dead components	18,011 lb, including connection allowance

Analysis and optimization method

PyNite 3D frame elements retain axial, bending, shear and torsion behavior. Shared columns are modeled once in the global system; support reactions are not arbitrarily divided in half between trusses. North-wall members and W4/N1/N2 remain coplanar at north coordinate 253 in.

Roof surface loads transfer to TE/TW through assumed east-west purlins at approximately 4 ft plan stations, with force and first moment preserved. The rear cap frame is a separate lightweight allowance. Those secondary members and their restraint connections still require design. Loft reactions transfer through the retained floor-beam/truss layout.

A 66-section AISC v16 catalog supplies published weight and geometric properties. B2 and the T1 floor rail remain W shapes at no more than 8 in depth. Existing 4 in column envelopes and the 6 in N1 envelope are retained. Thin HSS walls outside the compact-wall screening range are rejected.

For each candidate, section self-weight and global stiffness are updated, demands are recomputed, and strength/serviceability failures are rejected. The completed structure uses second-order P-delta analysis; the roof-first stage is linear and is not a complete erection-stability assessment.

The capacity routine screens axial compression buckling, tension, elastic biaxial bending, W-section lateral-torsional buckling and shear. HSS uses an additional conservative linear normal/bending/shear/torsion interaction. $K=1$ and a 72 in maximum restrained chord length are assumptions. This is a partial AISC-based screening routine, not a complete AISC 360 compliance engine.

Independent JSON reconstruction reproduced all 76 exported linear cases (56 complete, 20 roof-first). Maximum replay displacement difference was $8.24\text{e-}7$ in/radian; force difference 0.016 lb and moment difference 0.045 lb-in. A simple-beam benchmark matches its closed-form deflection. These validate implementation/replay, not the physical connection assumptions.

Serviceability and unresolved design checks

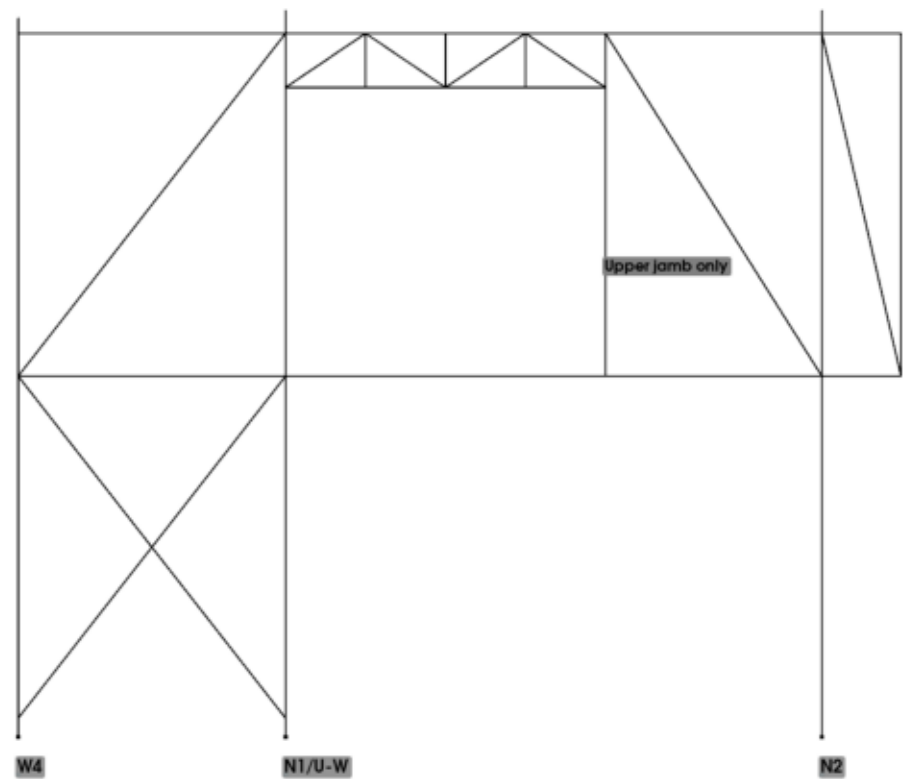
These are primary-frame study targets, not approval of the full floor, roof or hoist system. Floor joists, deck, vibration, glazing support and rail-manufacturer limits require their own checks.

The first pinned-base run tripped a near-zero-load equilibrium tolerance by 0.000037 lb. Repeating the actual service/strength combinations without the zero-total-gravity basic lateral cases solved successfully. It was not evidence of physical instability.

Check	Calculated	Study limit / outcome
Floor movement, live with bands	0.494 in	0.712 in (span/360)
Floor movement, total with hoist	0.749 in	1.068 in (span/240)
B2 incremental hoist movement	0.145 in	0.569 in study target
T1 incremental hoist movement	0.049 in	0.569 in study target
Roof total movement	0.240 in	1.054 in study target
Separate pinned-web screen	0.986 utilization	Passes implemented member screen
Separate pinned-base screen	0.987 utilization	Passes implemented member screen
Unrestrained chord sensitivity	8.44 utilization	FAIL: real chord restraint is necessary
125 psf storage sensitivity	1.60 utilization	FAIL: selected sizes do not cover this use
Wind, seismic, uplift	Sensitivity loads only	Site design and connections unresolved
Hoist and external jib crane	Vertical traveling hoist modeled	Rail flange/wheels unverified; jib not designed
Footings and connections	Not sized	No construction release

TN Sized truss elevation

Flat upper chord axis: 18 ft 8 1/4 in
Lower chord axis: 9 ft 7 in
Overall chord run: 23 ft 5 1/2 in



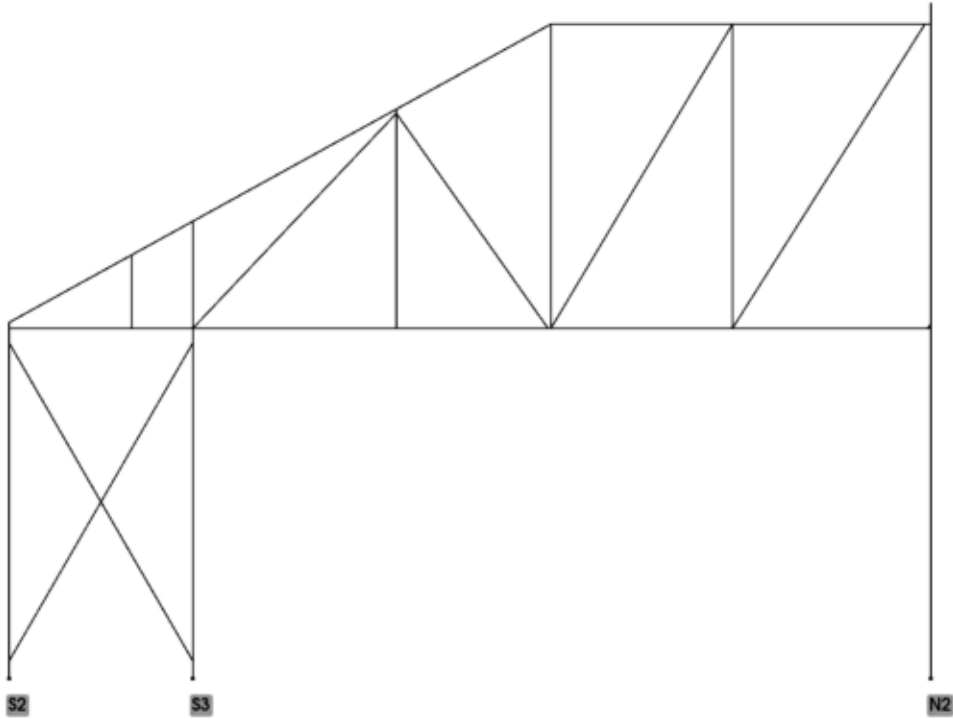
Fixed centerline geometry; selected stock sizes are listed in the member schedule.
Preliminary fixed-base model; connection and foundation design remains.

TN — member sizes

Utilization is the main completed/roof-first/P-delta envelope. Separate connection-sensitivity values are included in the CSV. Member labels identify physical pieces; lengths are centerline lengths, not cutting dimensions.

Member	Section	Length ft	Weight lb	Util.
T-N	HSS3X3X1/8	23.46	111.4	0.685
T-N upper east jamb	HSS4X4X3/16	9.10	85.8	0.110
T-N loading header	HSS1-1/2X1-1/2X1/8	8.50	18.7	0.432
T-N vertical 5	HSS1-1/2X1-1/2X1/8	9.10	20.0	0.063
T-N diagonal 8	HSS2X2X1/8	10.77	32.8	0.485
T-N diagonal 9	HSS1-1/2X1-1/2X1/8	9.34	20.6	0.144
T-N vertical 11	HSS1-1/2X1-1/2X1/8	1.44	3.2	0.039
T-N vertical 12	HSS1-1/2X1-1/2X1/8	1.44	3.2	0.027
T-N vertical 13	HSS1-1/2X1-1/2X1/8	1.44	3.2	0.036
T-N diagonal 15	HSS1-1/2X1-1/2X1/8	2.57	5.6	0.124
T-N diagonal 16	HSS1-1/2X1-1/2X1/8	2.57	5.6	0.085
T-N diagonal 17	HSS1-1/2X1-1/2X1/8	2.57	5.6	0.093
T-N diagonal 18	HSS1-1/2X1-1/2X1/8	2.57	5.6	0.128
T-N top / segment 1	HSS1-1/2X1-1/2X1/8	23.46	51.6	0.772

TE Sized truss elevation



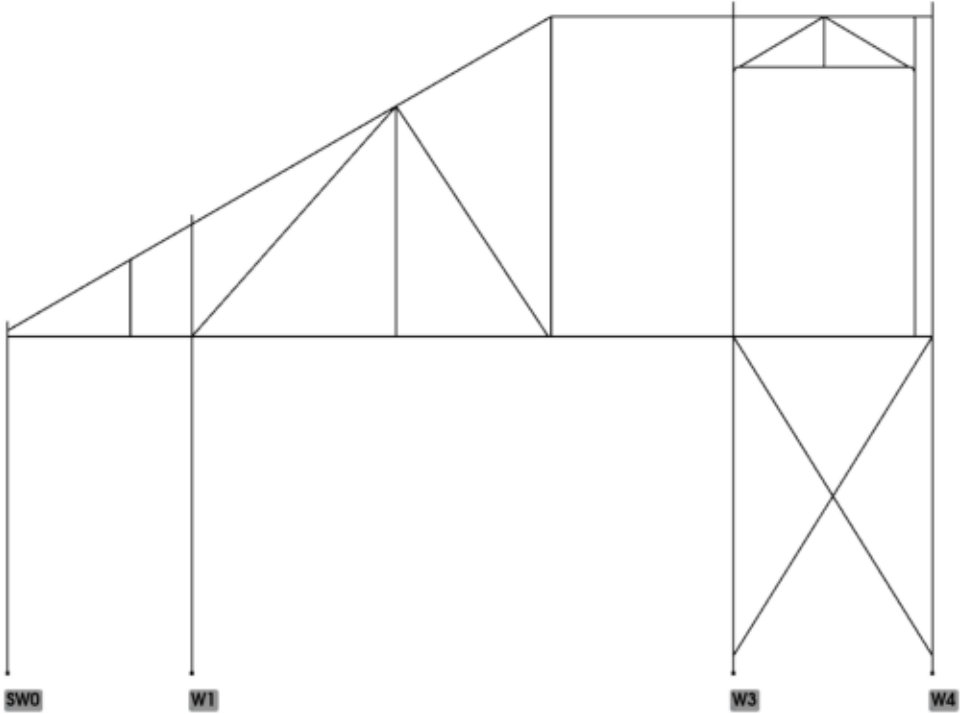
Fixed centerline geometry; selected stock sizes are listed in the member schedule.
Preliminary fixed-base model; connection and foundation design remains.

TE — member sizes

Utilization is the main completed/roof-first/P-delta envelope. Separate connection-sensitivity values are included in the CSV. Member labels identify physical pieces; lengths are centerline lengths, not cutting dimensions.

Member	Section	Length ft	Weight lb	Util.
T-E	HSS3X3X1/4	26.33	232.0	0.697
T-E vertical 1	HSS1-1/2X1-1/2X1/8	1.43	3.1	0.202
T-E vertical 3	HSS1-1/2X1-1/2X1/8	5.80	12.8	0.713
T-E vertical 4	HSS1-1/2X1-1/2X1/8	8.35	18.4	0.647
T-E vertical 6	HSS2X2X1/8	8.40	25.6	0.674
T-E diagonal 11	HSS2-1/4X2-1/4X1/8	8.45	29.4	0.727
T-E diagonal 12	HSS1-1/2X1-1/2X1/8	7.51	16.5	0.834
T-E diagonal 13	HSS2X2X1/8	10.28	31.4	0.784
T-E top / segment 1	HSS3X3X1/8	17.67	83.9	0.856
T-E top / segment 2	HSS4X4X3/16	10.85	102.2	0.823

TW Sized truss elevation



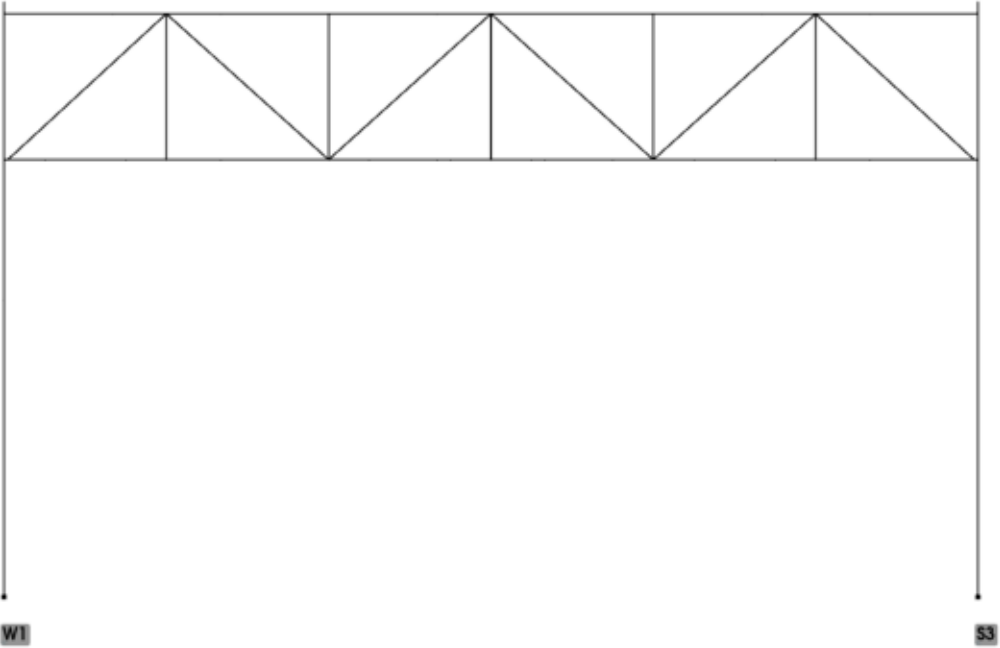
Fixed centerline geometry; selected stock sizes are listed in the member schedule.
Preliminary fixed-base model; connection and foundation design remains.

TW — member sizes

Utilization is the main completed/roof-first/P-delta envelope. Separate connection-sensitivity values are included in the CSV. Member labels identify physical pieces; lengths are centerline lengths, not cutting dimensions.

Member	Section	Length ft	Weight lb	Util.
T-W / B-WO	HSS3-1/2X3-1/2X3/16	26.33	214.6	0.794
T-W vertical 1	HSS1-1/2X1-1/2X1/8	1.76	3.9	0.178
T-W vertical 3	HSS1-1/2X1-1/2X1/8	6.13	13.5	0.666
T-W vertical 4	HSS1-1/2X1-1/2X1/8	8.68	19.1	0.316
T-W vertical 6	HSS1-1/2X1-1/2X1/8	8.73	19.2	0.208
T-W diagonal 9	HSS2-1/4X2-1/4X1/8	8.76	30.5	0.737
T-W diagonal 10	HSS1-1/2X1-1/2X1/8	7.86	17.3	0.753
T-W balcony header	HSS1-1/2X1-1/2X1/8	5.00	11.0	0.416
T-W vertical 13	HSS1-1/2X1-1/2X1/8	1.06	2.3	0.074
T-W diagonal 15	HSS1-1/2X1-1/2X1/8	2.81	6.2	0.175
T-W diagonal 16	HSS1-1/2X1-1/2X1/8	2.81	6.2	0.108
T-W top / segment 1	HSS2-1/2X2-1/2X1/8	17.88	69.7	0.830
T-W top / segment 2	HSS5X5X1/4	10.85	169.5	0.878

TS Sized truss elevation



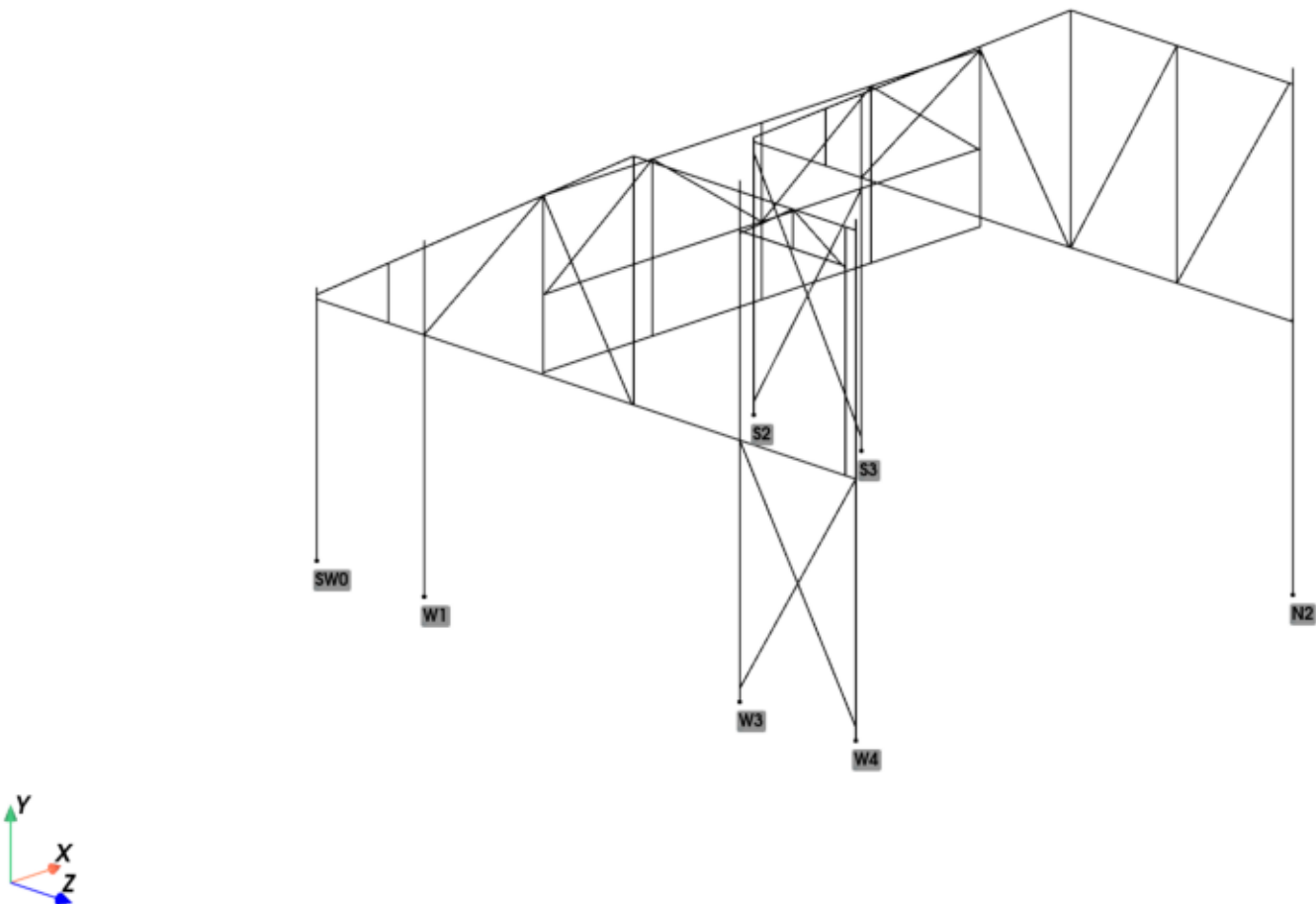
Fixed centerline geometry; selected stock sizes are listed in the member schedule.
Preliminary fixed-base model; connection and foundation design remains.

TS — member sizes

Utilization is the main completed/roof-first/P-delta envelope. Separate connection-sensitivity values are included in the CSV. Member labels identify physical pieces; lengths are centerline lengths, not cutting dimensions.

Member	Section	Length ft	Weight lb	Util.
T-S	HSS1-1/2X1-1/2X1/8	21.35	47.0	0.289
T-S vertical 2	HSS1-1/2X1-1/2X1/8	2.82	6.2	0.012
T-S vertical 3	HSS1-1/2X1-1/2X1/8	2.82	6.2	0.006
T-S vertical 4	HSS1-1/2X1-1/2X1/8	2.82	6.2	0.004
T-S vertical 5	HSS1-1/2X1-1/2X1/8	2.82	6.2	0.005
T-S vertical 6	HSS1-1/2X1-1/2X1/8	2.82	6.2	0.010
T-S diagonal 8	HSS1-1/2X1-1/2X1/8	4.72	10.4	0.088
T-S diagonal 9	HSS1-1/2X1-1/2X1/8	4.78	10.5	0.038
T-S diagonal 10	HSS1-1/2X1-1/2X1/8	4.78	10.5	0.047
T-S diagonal 11	HSS1-1/2X1-1/2X1/8	4.78	10.5	0.042
T-S diagonal 12	HSS1-1/2X1-1/2X1/8	4.78	10.5	0.041
T-S diagonal 13	HSS1-1/2X1-1/2X1/8	4.72	10.4	0.075
T-S top / segment 1	HSS1-1/2X1-1/2X1/8	21.35	47.0	0.158

T1 Sized support path



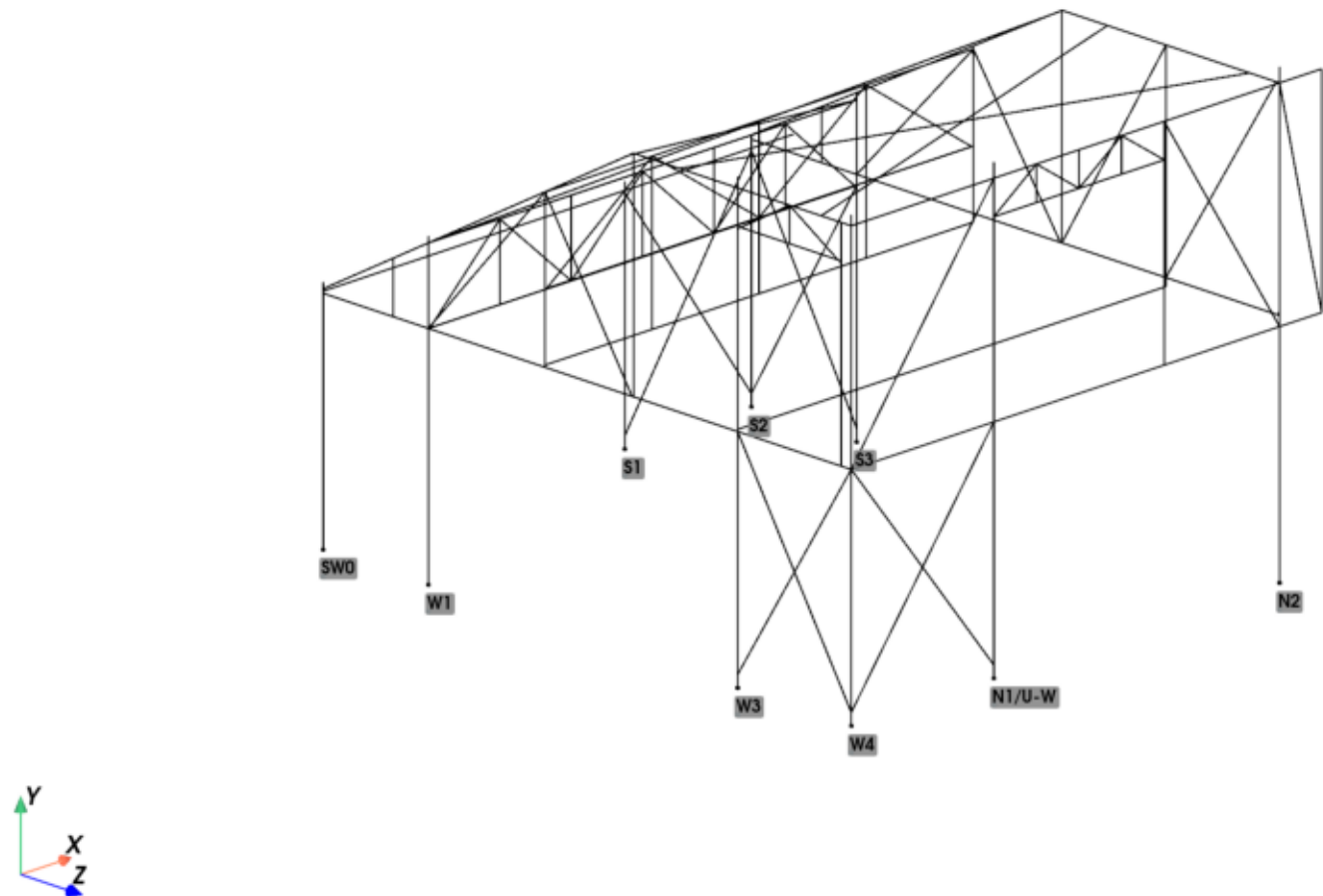
T1 bears on TE and TW; those trusses transfer its load to the ground columns.
No new posts beneath T1. This is a filtered 3D view of the globally solved frame.

T1 — member sizes

Utilization is the main completed/roof-first/P-delta envelope. Separate connection-sensitivity values are included in the CSV. Member labels identify physical pieces; lengths are centerline lengths, not cutting dimensions.

Member	Section	Length ft	Weight lb	Util.
T1 upper lower chord	HSS1-1/2X1-1/2X3/16	21.35	64.9	0.904
T1 future floor beam	W6X8.5	21.35	181.5	0.175
T1 vertical 2	HSS1-1/2X1-1/2X1/8	3.25	7.2	0.175
T1 vertical 3	HSS1-1/2X1-1/2X1/8	3.25	7.2	0.033
T1 vertical 4	HSS1-1/2X1-1/2X1/8	3.25	7.2	0.181
T1 diagonal 6	HSS1-1/2X1-1/2X3/16	6.38	19.4	0.890
T1 diagonal 7	HSS1-1/2X1-1/2X1/8	6.45	14.2	0.153
T1 diagonal 8	HSS1-1/2X1-1/2X1/8	6.45	14.2	0.138
T1 diagonal 9	HSS1-1/2X1-1/2X3/16	6.38	19.4	0.933
T1 future hanger 2	HSS1-1/2X1-1/2X1/8	2.26	5.0	0.333
T1 future hanger 3	HSS1-1/2X1-1/2X1/8	2.26	5.0	0.173
T1 future hanger 4	HSS1-1/2X1-1/2X1/8	2.26	5.0	0.281
T1 top / segment 1	HSS2X2X1/8	21.35	65.1	0.584

ALL Sized support path



Fixed centerline geometry; selected stock sizes are listed in the member schedule.
Preliminary fixed-base model; connection and foundation design remains.

ALL — member sizes

Utilization is the main completed/roof-first/P-delta envelope. Separate connection-sensitivity values are included in the CSV. Member labels identify physical pieces; lengths are centerline lengths, not cutting dimensions.

Member	Section	Length ft	Weight lb	Util.
W1	HSS4X4X3/16	13.03	122.8	0.308
W3	HSS4X4X1/2	19.09	412.9	0.869
W4	HSS4X4X1/4	19.09	233.1	0.822
S1	HSS4X4X3/16	10.00	94.2	0.071
S2	HSS4X4X3/16	10.00	94.2	0.563
N2	HSS4X4X1/4	19.28	235.4	0.749
S3	HSS4X4X3/16	13.03	122.8	0.685
SW0	HSS4X4X3/16	10.00	94.2	0.179
T-SO	W6X8.5	23.46	199.4	0.138
B2 future floor beam	W8X24	21.35	512.5	0.498
N1 / U-W	HSS6X6X1/4	19.29	366.9	0.309

Bracing — member sizes

These physical braces are included in primary steel weight. Chord lateral restraint at 6 ft remains an additional required secondary system, not implied by the presence of these braces.

Member	Section	Length ft	Weight lb	Util.
BR-W-1	HSS1-1/2X1-1/2X1/8	10.71	23.6	0.350
BR-W-2	HSS1-1/2X1-1/2X1/8	10.71	23.6	0.692
BR-S-1	HSS1-1/2X1-1/2X1/8	11.06	24.3	0.396
BR-S-2	HSS1-1/2X1-1/2X1/8	11.06	24.3	0.254
BR-R-main-1	HSS2-1/4X2-1/4X1/8	24.28	84.5	0.789
BR-R-main-2	HSS2X2X1/8	24.28	74.0	0.841
BR-R-cap-1	HSS3X3X3/16	22.50	154.6	0.854
BR-R-cap-2	HSS1-1/2X1-1/2X1/8	22.50	49.5	0.642
BR-E-upper-1	HSS2X2X1/8	10.12	30.9	0.526
BR-N-upper-1	HSS2-1/2X2-1/2X1/8	11.55	45.0	0.805
BR-N-ground-1	HSS2X2X1/8	11.53	35.2	0.951
BR-N-ground-2	HSS2-1/4X2-1/4X1/8	11.53	40.1	0.838
BR-E-ground-1	HSS2-1/2X2-1/2X1/8	10.49	40.9	0.711
BR-E-ground-2	HSS2-1/4X2-1/4X1/8	10.49	36.5	0.852

Sources and how to use this study

Canadian Solar TOPHiKu6 CS6.1-54TM-H datasheet (2026): 50.7 lb over about 21.96 sq ft, or about 2.31 psf for the module alone. The study reserves 3 psf for PV or matching infill glass; actual cut/infill panels may differ.

Metl-Span CF/PIR and CFR panel weight tables inform the 3 psf insulated metal-skin budget. Profile, gauge, insulation and attachment choices must be confirmed. The floor and secondary-steel allowances are preliminary engineering budgets.

Sources are linked below and recorded as full URLs in load-basis.json. San Diego occupancy classification and site lateral/environmental loads must be resolved before the selected sizes can become a construction design.

Use member-schedule.csv for each member and column; column-reactions.csv for exported linear load-case base actions; global-complete-PDelta.json for second-order results; complete.network.json and roof_first.network.json for a rebuildable analysis model. Interactive HTML views are native solver scenes.

Next phase: first establish physical chord restraint, connections and trolley compatibility; then compare alternative layouts and fabrication cost. A lighter primary member can require heavier secondary bracing or more fabrication, so lowest primary weight is not necessarily lowest installed cost.

Primary references

PV

[Open official source](#)

PANEL

[Open official source](#)

ROOF_PANEL

[Open official source](#)

CODE

[Open official source](#)

SECTIONS

[Open official source](#)

AISC capacity implementation: partial preliminary screen; see optimize.py and iterate_design.py.